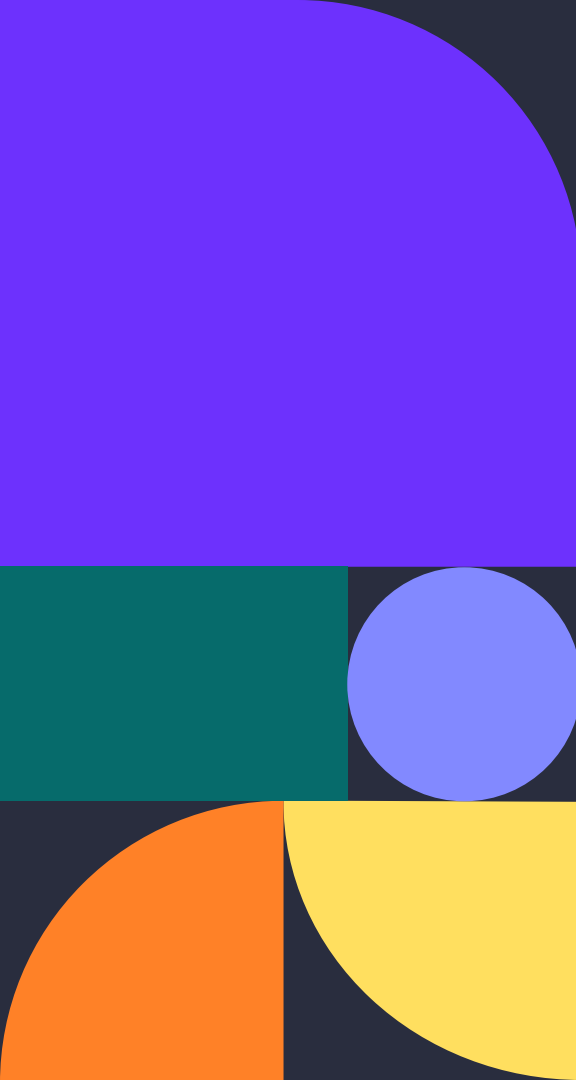
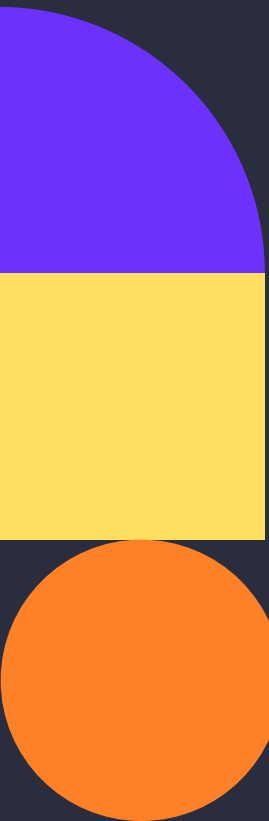




Cybersecurity Ethics: Theory & Applicability

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Learning Objectives

01. **Explore current Cybersecurity standards**
02. **Understand their theoretical backgrounds**
03. **Understand the core principles of cybersecurity ethics**
04. **Analyze controversial case studies**



Current Cybersecurity Ethics

There are none! :)

...Officially that is.

Organizations such as academic institutes, companies, and nonprofits have their own codes of ethic geared towards cybersecurity professionals.

There is no distinct "single ethical guideline" that professionals are expected to follow like there is for more technical aspects of the profession.



Personal Opinion: Best resources for Cybersecurity Ethics Right Now:

- ACM Code of Ethics:
<https://www.acm.org/code-of-ethics>
- Santa Clara University Cybersecurity Ethics Courses:
<https://www.scu.edu/media/ethics-center/technology-ethics/IntroToCybersecurityEthics.pdf>

Most Occuring Standards

Honesty

- Transparency in capabilities of work and qualifications
- No misleading claims
- No false data
- No bribes
- Mention any conflicts of interest
- No misrepresenting organizations

Confidentiality/Privacy

- Respect the privacy of users as much as you can
- Be transparent in the data you collect
- Use the minimum amount of personal information necessary
- Protect client data such as trade secrets

Improve Society

- Your work should aim to improve computing for all of society
- Minimize the harm you do

Maintain Professional Knowledge

- Maintain professional technical skills
- Grow as the technical landscape does
- Recognize areas that need work



Aside:

What about the law?

- Obviously, FOLLOW YOUR LOCAL LAWS!!!
- However, cybersecurity ethics have a strong key flaw in globality
- The internet and tech is global
- Laws are not globally applicable
- Codes of ethic are either bound to their locality or have vague clauses of "follow the law"
- This leaves people unprotected, and can lead to variability in how professionals should handle situations



Theoretical Backgrounds



Aristotle

Nicomachean Ethics

- Core western ethical document
- Has ties to most contemporary ethical theories
- Written around 335 BCE
- Book IV focuses on some of the "moral virtues"
- Book VI focuses on "intellectual virtues"
- Can indirectly tie to all of the standards
- Directly ties to "honesty" and "maintaining professional knowledge"



Immanuel Kant

Categorical Imperative

- Introduced in "The Groundwork of the Metaphysics of Morals" in 1785
- "Categorical Imperative" is maxims that must be applied universally at all times to all people.
- Introduces the idea that people are "ends within themselves"
- Indirectly connects to all the standards
- Is groundwork for the next theoretical background





John Stuart Mill: Utilitarianism

- Written in 1861
- The most common ethical framework cited in our current contemporary moment
- The very tldr of the idea is actions should be done to increase the most pleasure possible in the world and reduce the most pain
- Applies to nearly every cybersecurity standard, and is possibly the source for some of them like the “improve society” standard



Looking Forward:

How can we better develop cybersecurity ethics?

- If we examine the theoretical backgrounds for our standards, we can find new ways to critique the underlying reasoning for the standard
- Once we rethink the theory, we can create standards that better fit our needs.
- People don't agree on ethics nearly ever, and that's why we grow
- You have to follow the ethics that people expect of you right now, but you can always work to change those for the future

Case Studies

Alright we are going to look at some case studies.

Each one you will have time to turn and talk with your partner.

Then we can discuss as a class after you bounce ideas around.



Case Study: Bossware

- In recent years, companies have adopted "Bossware"—monitoring software installed on employee devices—to track productivity, screen time, and even keystrokes. Some versions enable webcam activation and screen recording. With remote work on the rise, Bossware can be used in employees' homes.
- Installed with or without employee knowledge
- Often used on personal or BYOD (Bring Your Own Device) setups
- Raises concerns when applied to employees' private spaces or family devices
- Parallels are drawn to parental control software on children's devices

Case Study: Hello Barbie

- In 2015, Mattel released Hello Barbie, a Wi-Fi-connected doll that used voice recognition and cloud storage to allow children to have interactive conversations. Parents could access recordings of their children's speech.
- Collected and stored children's voices on remote servers
- Security researchers found major vulnerabilities
- Concerns about surveillance and informed parental consent emerged
- Critics raised alarms over commercialization of child data

Case Study:

Student Data Monitoring

- The University of Arizona conducted research analyzing student ID card swipe data to predict class attendance, routines, and social networks. The intent was to identify at-risk students and prevent dropouts.
- No formal consent process from students
- The data included library visits, dining hall entries, and dorm access
- Findings were used to shape academic interventions
- Debate arose over data privacy, transparency, and academic benefit

Case Study: Edward Snowden

- In 2013, Edward Snowden, a contractor for the NSA, leaked classified documents revealing global surveillance programs by U.S. and allied governments. The programs included bulk collection of phone records and internet activity.
- Snowden fled to Hong Kong and then Moscow
- He claimed whistleblower status; others labeled him a traitor
- The U.S. charged him under the Espionage Act
- The leaks sparked global discussions on surveillance and privacy

Any questions? Ask away!

I am happy to try to answer what I can, but I can't promise an answer for every question.



Thank you